



Short Communication

Genomic insights of *optrA*-carrying linezolid-resistant *Enterococcus faecium* using hybrid assembly: first report from India

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ABSTRACT

Objectives: Linezolid resistance in *Enterococcus faecium* is emerging worldwide. In this study, we aimed to characterise two linezolid-resistant *E. faecium* isolates using whole-genome sequencing.

Methods: Antimicrobial susceptibility testing was performed by the broth microdilution method. A hybrid assembly approach of IonTorrent and MinION sequencing reads was used to generate the complete genome of linezolid-resistant *E. faecium* isolates VB3025 and VB3240.

Results: VB3025 and VB3240 had minimum inhibitory concentration (MICs) for linezolid of 1024 µg/mL and 512 µg/mL, respectively. In addition, VB3025 was found to be resistant to both vancomycin and teicoplanin, while VB3240 was susceptible to these antibiotics. A hybrid assembly approach was used to generate the complete genome of VB3025 and VB3240 isolates harbouring the *optrA* gene. Notably, VB3025 carried two copies of *optrA* (chromosomal and plasmid), while in VB3240 *optrA* was identified on the chromosome. Interestingly, the plasmid pVB3025_2 co-carried the resistance gene clusters *aph(3)-IIIa-sat4-ant(6)-la-ermB*, the *vanHAX* operon and a copy of the *optrA* gene. Moreover, the *optrA* gene inserted into a Tn554 transposon carrying the *ermA* gene was identified in both VB3025 and VB3240 isolates. Furthermore, mutation analysis revealed the presence of a G2592T mutation in the 23S rRNA of both isolates.

Conclusion: This is the first study reporting *optrA*-positive linezolid-resistant *E. faecium* from India. A novel plasmid co-carrying vancomycin and linezolid resistance determinants highlights the threat for potential dissemination.

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1. Introduction

Enterococcus faecium infections are often difficult to treat owing to intrinsic and acquired resistance to multiple antibiotics. Linezolid is being increasingly used for the treatment of infections caused by vancomycin-resistant enterococci [1]. Linezolid inhibits protein synthesis by binding to domain V of 23S rRNA of the 50S ribosomal subunit. Resistance to linezolid can be mediated either by a G2576T mutation in domain V of 23S rRNA or by the acquisition of *cfr*, *optrA* and *poxA* genes [2]. The risk of nosocomial spread of linezolid resistance in enterococci is currently of significant clinical concern.

However, the incidence of linezolid resistance in enterococci remains low (<1%) across the world (ATLAS database; <https://atlas-surveillance.com>).

Among the transferable linezolid resistance determinants, *optrA* has been frequently reported and is responsible for the recent increase in linezolid-resistant enterococci (LRE) among clinical isolates [2–7]. The *optrA* gene encodes an ATP-binding cassette F (ABC-F) ribosomal protection protein conferring cross-resistance to phenicols and oxazolidinones, including tedizolid [7]. Several studies have reported LRE in India [8–10]. The emergence of LRE following short-term linezolid therapy has also been reported with the occurrence of G2576T mutation in 23S rRNA of *E. faecium* [10]. However, studies investigating the genomics of LRE are not available from India. In the present study, we aimed to investigate the genomes of two linezolid-resistant *E. faecium* isolates using a

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hybrid assembly approach of IonTorrent and MinION sequencing reads.

2. Materials and methods

2.1. Bacterial isolates

During January–December 2019, we identified two sporadic cases of linezolid-resistant *E. faecium* in patients admitted to Christian Medical College (Vellore, India) at different time points and from different states of India. *Enterococcus faecium* isolate VB3025 was recovered from a blood culture of a patient from Tamil Nadu (admitted in July 2019) with stage IV left ischial ulcers of the left lower limb and fever for 10 days. *Enterococcus faecium* isolate VB3240 was recovered from a blood culture of a patient involved in a road traffic accident with crush injury and a fracture of the right arm and forearm who was treated elsewhere in West Bengal and then referred to Christian Medical College (in September 2019) for further management. Identification of *E. faecium* was carried out using standard microbiological methods [11]. Furthermore, the isolates were re-confirmed as *E. faecium* using matrix-assisted laser desorption/ionisation time-of-flight mass spectrometry (MALDI-TOF/MS) (VITEK® MS; bioMérieux, Marcy-l'Étoile, France).

2.2. Antimicrobial susceptibility testing

Minimum inhibitory concentrations (MICs) of ampicillin, erythromycin, levofloxacin, tetracycline, chloramphenicol, minocycline, vancomycin, linezolid and tigecycline were determined by the broth microdilution method. Interpretation was carried out according to Clinical and Laboratory Standards Institute (CLSI) guidelines [12], except for tigecycline which was interpreted according to European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines [13].

2.3. Genome sequencing and assembly

2.3.1. DNA extraction

Total genomic DNA of *E. faecium* isolates VB3025 and VB3240 was extracted from an overnight culture (14–16 h) grown at 37°C on blood agar using a QIAamp® DNA Mini Kit and a QIAcube instrument (QIAGEN, Valencia, CA, USA) according to the manufacturer's instructions. The DNA concentration was measured using a Qubit fluorometer and the DNA was stored at –20°C.

2.3.2. Genome sequencing and assembly

Briefly, short-read sequencing was performed with 400-bp read chemistry using an IonTorrent Personal Genome Machine™ (PGM) (Life Technologies, Carlsbad, CA, USA) according to the manufacturer's instructions. For long-read sequencing, a MinION sequencing device was used with FLO-MIN106 R9 MinION flow cells (Oxford Nanopore Technologies, Oxford, UK). A long-read DNA library was prepared using an SQK-LSK108 Ligation Sequencing Kit v.R9 along with an ONT EXP-NBD103 Native Barcode Expansion kit (Oxford Nanopore Technologies) following the manufacturer's protocol. The library was loaded onto the flow cells and was run for 48 h using the standard MinKNOW software.

To obtain the complete genome, hybrid de novo assembly, quality checking and associated assembly statistics were performed using Illumina and ONT MinION reads as described previously [14]. Fast5 files were generated from MinION sequencing and the reads were base-called using Albacore 2.0.1 (<https://nanoporetech.com/about-us/news/new-basecaller-now-performs-raw-basecalling-improved-sequencing-and-assembly>)

and the adapter sequences were trimmed off using Porechop (<https://github.com/rrwick/Porechop>). The quality of the MinION reads was assessed using MinIONQC (https://github.com/roblanf/minion_qc). For accurate assembly, the MinION long reads were error-corrected with the standalone Canu v.1.7 error correction tool using the 'correct-nanopore-raw' module [15]. This was followed by hybrid de novo assembly using both IonTorrent and MinION reads by means of the Unicycler hybrid assembly pipeline v.0.4.6 for prokaryotic genomes with the default settings [16]. By default, Unicycler utilises SPAdes [17] to assemble the short reads with different *k*-mers and to filter out the low-depth regions along with error correction and quality checks. Subsequently, it trims and generates the short-read assembly graph. In addition, it uses miniasm [18] and Racon [19] to assemble the MinION long reads and further to bridge the reads to determine all the genome repeats and to produce a complete genome assembly. The complete circular genome was initially assembled and was polished with multiple rounds of Pilon v.1.22 to reduce the base level errors as described previously [20]. The final assembly was further polished with at least two or three rounds of Nanopolish v.0.7.1 (<https://github.com/jts/nanopolish>) to reduce the presence of indels.

2.4. Genome annotation and downstream analysis

The genome was annotated using the Prokaryotic Genome Annotation Pipeline (PGAP) v.4.1 (https://www.ncbi.nlm.nih.gov/genome/annotation_prok/). Sequence types (STs) of the isolates were determined using the PubMLST database of *E. faecium* (<https://pubmlst.org/efaecium/>). Furthermore, virulence factors were identified using VirulenceFinder 2.0 (<https://cge.cbs.dtu.dk/services/VirulenceFinder/>). Similarly, the presence of antimicrobial resistance genes was identified using ResFinder 4.1 (<https://cge.cbs.dtu.dk/services/ResFinder/>). In addition, the presence of plasmids in the genome was identified and characterised using PlasmidFinder 2.1 (<https://cge.cbs.dtu.dk/services/PlasmidFinder/>). The genomes of VB3025 and VB3240 were also screened for the presence of mutations in 23S rRNA and in the ribosomal protein genes *rplC*, *rplD* and *rplV*.

3. Results

3.1. Antimicrobial susceptibility testing

Phenotypic antimicrobial susceptibility testing of isolates VB3025 and VB3240 showed resistance to ampicillin, high-level gentamicin, tetracycline, minocycline, chloramphenicol and linezolid (Table 1). Isolate VB3025 was found to be resistant to both vancomycin and teicoplanin, while VB3240 was susceptible to these antibiotics. However, both isolates (VB3025 and VB3240) were susceptible to tigecycline (MIC = 0.12 µg/mL).

3.2. Genome features, virulence factors and resistance determinants

Whole-genome sequencing was performed and hybrid assemblies were generated to obtain complete chromosome and plasmid structures. The genome of *E. faecium* isolate VB3025 contained a chromosome (2.85 Mb) and two distinct plasmids. The genome of *E. faecium* isolate VB3240 carried a chromosome (2.80 Mb) and five different plasmids. Isolates VB3025 and VB3240 belonged to ST80 and ST17, respectively. In addition, genome analysis revealed the presence of adhesin-encoding genes such as *acm*, *efafm* and *scm* in both isolates. Furthermore, antimicrobial resistance genes were identified in the genomes (chromosome and plasmids) of VB3025 and VB3240 and are listed in Table 1.

Table 1Antimicrobial resistance genes (ARGs) identified in the genomes of *Enterococcus faecium* isolates VB3025 and VB3240

Antibiotic	MIC (μ g/mL) or zone diameter (mm) VB3025	MIC (μ g/mL) or zone diameter (mm) VB3240	ARGs and mutations	VB3025 (ST80) Chromosome	pVB3025_1	pVB3025_2	VB3240 (ST17) Chromosome	pVB3240_1
Ampicillin	> 32	> 32	Mutations in <i>pbp5</i> ^a	+	–	–	+	–
High-level gen-tam-icin	No zone of inhibition	No zone of inhibition	<i>aac(6')-li</i>	+	–	–	+	–
Streptothricin	Not tested	Not tested	<i>aac(6')-aph(2'')</i>	–	+	–	–	+
Vancomycin (disk)	> 32	16	<i>aph(3)-IIIa</i>	–	–	+	–	–
			<i>ant(6)-Ia</i>	–	–	+	–	–
			<i>aph(2'')-Ia</i>	–	–	–	–	+
			<i>sat4</i>	–	–	+	–	–
			<i>ermA</i>	+	–	+	+	–
			<i>ermB</i>	–	+	+	–	–
			<i>msrC</i>	+	–	–	+	–
			<i>ermT</i>	–	–	–	+	–
Levofloxacin	8	16	<i>gyrA</i> (S83Y)	+	–	–	+	–
			<i>parC</i> (S80I)	+	–	–	–	–
			<i>parC</i> (E84K)	–	–	–	+	–
Tetracycline, minocycline	16	> 32	<i>tetS</i>	–	–	+	–	–
			<i>tetL</i>	–	–	–	+	–
			<i>tetM</i>	–	–	–	+	–
Vancomycin/teicoplanin	> 256 / > 256	2 / 4	<i>vanA</i>	–	–	+	–	–
Linezolid	1024	512	23S rRNA (G2592T)	+	–	–	+	–
			<i>optrA</i>	+	–	+	+	–
			<i>rplC</i> (K158E)	–	–	–	+	–
Chloramphenicol	512	64	<i>optrA</i> gene exhibits cross-resistance to chloramphenicol and linezolid					

MIC, minimum inhibitory concentration; ST, sequence type.

^a Mutations identified in *pbp5* of VB3025 and VB3240: V24A, S27G, R34Q, G66E, A68T, E85D, E100Q, K144Q, T172A, L177I, D204G, A216S, T324A, M485A, N496K, A499T, E525D, E629V and P667S.^b *tetS* and *tetM* confer resistance to both tetracycline and minocycline.

3.3. Genetic environment of the *optrA* gene

A Tn554 transposon-mediated insertion of the *optrA* gene was identified in the chromosome both of VB3025 and VB3240 as well as in plasmid pVB3025_2 (Fig. 1). Moreover, transposon Tn554 was inserted into the *radC* gene (encoding a DNA repair protein) in the chromosomes of VB3025 and VB3240 and in plasmid pVB3025_2. The transcriptional regulator gene *araC* upstream and an *ermA* gene downstream of *optrA* were identified (Fig. 1).

3.4. Plasmids in isolate VB3025

In VB3025, two novel plasmids (pVB3025_1 and pVB3025_2) were identified. An aminoglycoside resistance-encoding gene *aac(6')-aph(2'')* and a macrolide resistance determinant *ermB* were identified on plasmid pVB3025_1. BLAST match analysis of plasmid pVB3025_1 shared 99.9% identity with a query coverage of 80% to the *E. faecium* AUSMDU00004142 plasmid (GenBank accession no. **CP027502**).

Multiple resistance genes were identified on plasmid pVB3025_2. A Tn3-like transposon-mediated insertion of the multiresistance gene cluster *aph(3)-IIIa-sat4-ant(6)-Ia-ermB* (macrolide-aminoglycoside-streptothricin element) was observed. In addition, a *vanHAX* operon associated with transposon Tn1546 and flanked by an insertion sequence IS1216 was found. Moreover, *optrA* co-localised with *ermA* on a Tn554 transposon was identified on plasmid pVB3025_2 (Fig. 2). BLAST match analysis of pVB3025_2 showed 96.9% similarity with a query coverage of 50% to *E. faecium* E6988 plasmid 3 (GenBank accession no. **LR135245**) and 96.1% similarity with a query coverage of 70% to *E. faecium* strain SC4 plasmid p2 (GenBank accession no. CP025427). In addition, an integrated plasmid was identified in the chromosome of VB3025 that exhibits 99.4% similarity with a query coverage of 100% to plasmid pHVH-V1836-3 (GenBank accession no. **CP044267.1**).

3.5. Plasmids of isolate VB3240

Plasmid pVB3240_1 carried an aminoglycoside resistance gene *aac(6')-aph(2'')* and two copies of *aph(2'')-Ia*. However, none of the remaining plasmids (pVB3240_2,3,4,5) carried either antimicrobial resistance or virulence determinants. BLAST analysis of pVB3240_1 exhibited 99.9% similarity with a query coverage of 86% to *E. faecium* SC4 plasmid p1 (GenBank accession no. **CP025426.1**). Similarly, plasmid pVB3240_2 showed 92.96% similarity with a query coverage of 100% to *Enterococcus faecalis* plasmid pCLF1 (GenBank accession no. **KU892126.1**). Plasmid pVB3240_3 exhibited 100% similarity and query coverage to *E. faecium* E8172 plasmid 12 (GenBank accession no. **LR135383.1**). Furthermore, plasmid pVB3240_4 displayed 99.7% identity with a query coverage of 100% to *E. faecium* E7114 plasmid 6 (GenBank accession no. **LR135448.1**), and plasmid pVB3240_5 shared 99% similarity and query coverage to *E. faecium* E8195 plasmid 7 (GenBank accession no. **LR135370.1**).

4. Discussion

Linezolid remains a valuable treatment option for the management of vancomycin-resistant enterococcal infections. Here we present a genomic investigation of two *optrA*-positive *E. faecium* isolates (VB3025 and VB3240). To the best of our knowledge, this is the first study reporting *optrA*-carrying *E. faecium* from India. However, linezolid resistance in *E. faecium* is not encountered in this hospital setting except for these two sporadic cases.

The G2576T mutation in 23S rRNA is the most frequent mechanism of resistance in LRE [21]. In 2015, *optrA* was first reported in *E. faecalis* of human origin from China [7]. Subsequently, a national surveillance study from China reported that 2% of enterococci isolates were *optrA*-positive [6]. Of the two species, *E. faecalis* is more often isolated with the *optrA* gene than *E. faecium* [3]. A study has reported that *optrA*-positive enterococci are more frequently encountered in animals [7] and are now emerging in human infections [2].

In addition to the plasmid-borne *optrA* gene [4,22,23], transposon-mediated insertion of the *optrA* gene into the chromo-

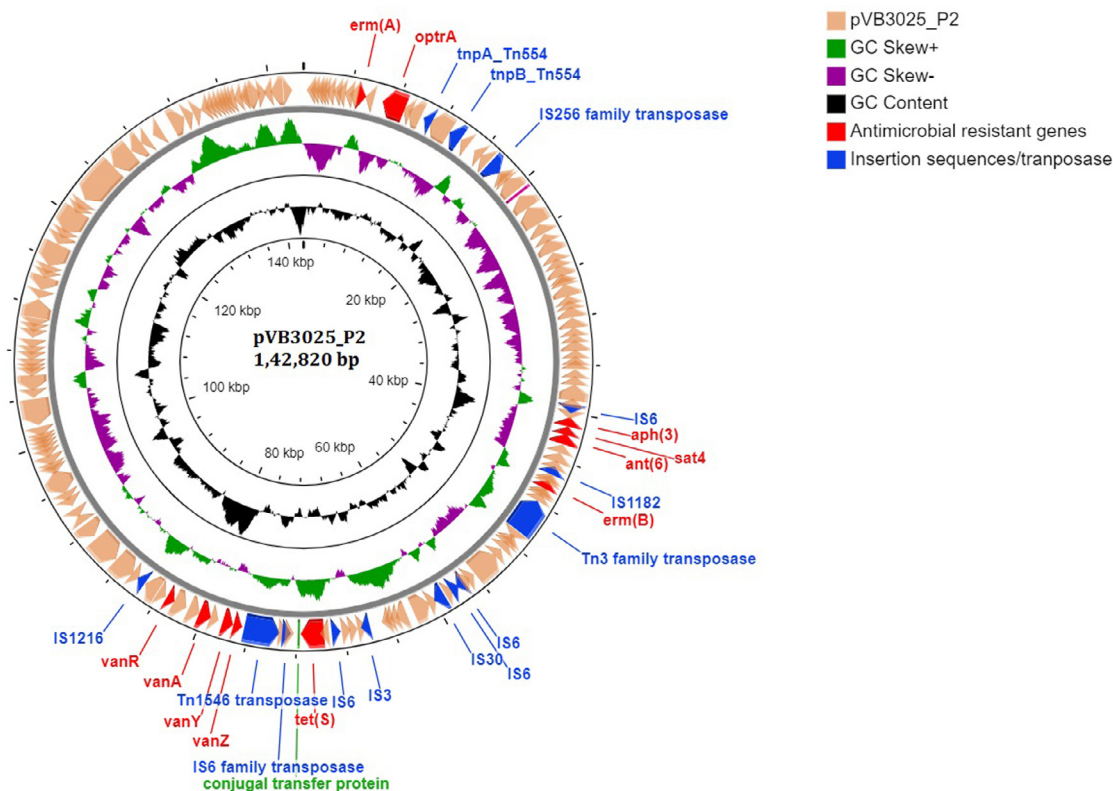


Fig. 1. Genetic environment of the *optrA* gene identified in *Enterococcus faecium* isolates VB3025 and VB3240. The Tn554 transposon carrying *optrA* and its flanking region of VB3025 were aligned against VB3240 and plasmid pVB3025_2 using BLASTn algorithm and similar regions are represented with >80% identity. The orientation of coding regions is indicated by arrows. Transposase are indicated in yellow and antimicrobial resistance genes are in red.

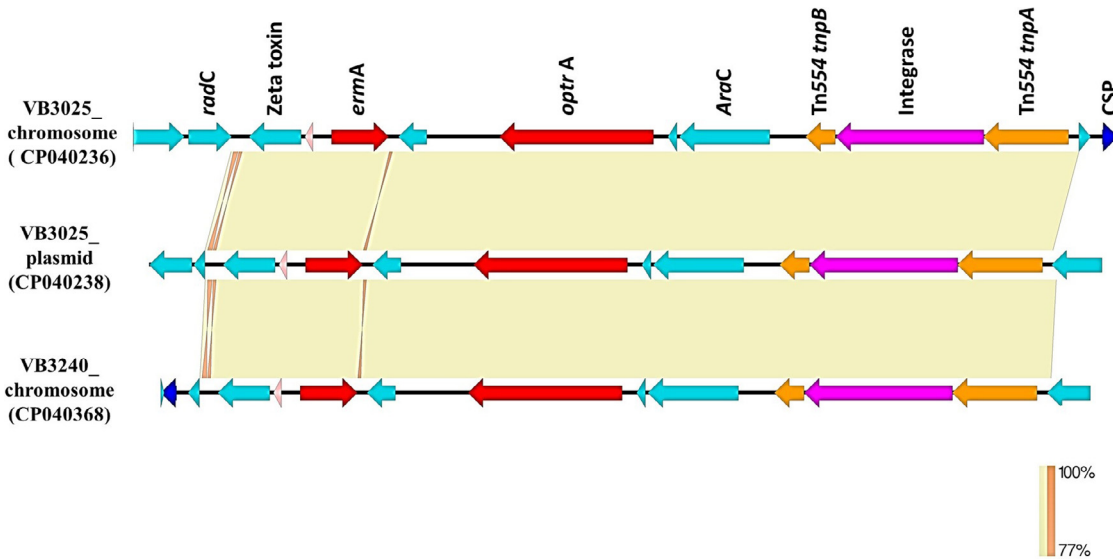


Fig. 2. Circular genomic representation of plasmid pVB3025_P2. The location of antimicrobial resistance genes is denoted in red and insertion sequence (IS) elements and transposases are denoted in blue.

some has also been documented [24,25]. A novel MDR transposon Tn6674 belonging to the Tn554 transposon family carrying the gene cluster *spc-ermA-fexA-optrA* encoding resistance to spectinomycin, macrolides, phenicols and oxazolidinone has reported in *E. faecalis* [24]. Furthermore, the Tn6674 transposon shares a similar genetic structure to that of Tn554, which carries the resistance genes *spc* and *ermA* [24]. Similarly, in the present study, the *optrA* gene was found to be inserted into a Tn554 transposon carry-

ing the *ermA* gene. However, in the present study, all of the identified Tn554 transposons lacked the *spc* (spectinomycin adenyltransferase) gene that has been previously described [4]. The *radC* gene has been demonstrated to be the common integration site for Tn554 family transposons [26]. Concurrently, in the present study, Tn554-mediated insertion of resistance determinants (*ermA* and *optrA*) was found in the chromosome both of isolates VB3025 and VB3240 and in plasmid pVB3025_2.

Studies have also reported the co-occurrence of *optrA* with the *cfr* gene [22,27] and *optrA* with the *poxA* gene on plasmids [28,29]. Although studies describing *optrA* in vancomycin-resistant enterococci (located on different plasmids) are common, this is the first study to report the co-occurrence of *optrA* with the *vanHAX* operon on the same plasmid (pVB3025_2). BLAST analysis of plasmid pVB3025_2 showed 53% identity to that of the *E. faecium* UW7931 plasmid carrying *optrA* with the *aadE-sat4-aphA-3* gene cluster [30]. Location of the *optrA* gene on the chromosome or in the plasmid, or both, has been reported [3]. Interestingly, in the present study, two copies of *optrA* (one on the chromosome and another on plasmid pVB3025_2) were identified in the isolate VB3025. In addition to the *optrA* gene, a G2592T mutation was also identified in the 23S rRNA both of isolates VB3025 and VB3240. This distinctive combined mechanism of linezolid resistance has not been reported elsewhere. The *optrA* gene is associated with diverse sequence types (ST30, ST78, ST80, ST168, ST273, ST650, ST927, ST16, ST17 and ST80) and is carried on various plasmids [4,31,32]. However, these STs or plasmid types have not yet been associated with *optrA* on a global scale. Moreover, plasmids identified in the study isolates are novel and have not been reported previously.

In conclusion, the novelty of the present study findings is the co-localisation of *optrA* and the *vanHAX* operon on plasmid pVB3025_2. In addition to *optrA*, a G2592T mutation in domain V of 23S rRNA was also identified both in isolates VB3025 and VB3240. Transposon Tn554-mediated insertion of resistance determinants (*ermA* and *optrA*) in the chromosome of VB3025 and VB3240 and in plasmid pVB3025_2 were observed. This observation reiterates that Tn554 may promote the dissemination of the *optrA* gene. Notably, the co-existence of the vancomycin resistance-encoding *vanHAX* operon with a copy of the *optrA* gene on a novel plasmid (pVB3025_2) highlights the threat for potential rapid dissemination.

Declaration of Competing Interest

None declared.

Data availability

The genome sequences of the chromosomes and plasmids have been deposited in GenBank under the following accession numbers: VB3025 (chromosome **CP040236**; plasmids **CP040237** and **CP040238**); and VB3240 (chromosome **CP040368**; plasmids **CP040369**, **CP040370**, **CP040371**, **CP040372** and **CP040373**).

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Ethical approval

Not required.

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